TEXORA · LECTURE SHEET

T-test

Student's t distribution and its properties, the one-sample t-test for a mean with a fully worked breaking-load example, the paired t-test for before-and-after measurements, and the pooled t-test for two independent samples.

STAT-2109 · PROBABILITY AND STATISTICS · TOPIC 10 · LEVEL 2 TERM 1

1. Why small samples need a new distribution

The z-test of Topic 9 rests on two comforts small samples do not provide: a large sample, so that the central limit theorem makes the sample mean effectively normal, and a known population standard deviation σ. In laboratory and mill work neither is available: a breaking-load test may use only 8 or 10 specimens, and σ must be estimated from the same small sample through s. Replacing σ by s adds uncertainty, because s itself varies from sample to sample. W. S. Gosset, writing as "Student", worked out the exact distribution allowing for this; the resulting **t-test** is the standard tool for a mean when the sample is small and σ is unknown. Its logic is that of Topic 9 — hypotheses, level, statistic, computation, decision — but the statistic is referred to the t distribution, and the sample size enters through the degrees of freedom.

2. Student's t distribution: properties

- **Symmetric and bell shaped.** Like the standard normal curve, the t curve is symmetric about zero, so positive and negative departures of the same size are equally likely under the null hypothesis.
- **Heavier tails.** The t curve is flatter at the centre and thicker in the tails than the normal curve. Extreme values are more likely, which is the distribution's way of allowing for the uncertainty in s. In consequence, t critical values are larger than the corresponding z values: stronger evidence is demanded before rejection.
- **Degrees of freedom.** There is not one t curve but a family of them, indexed by the **degrees of freedom**, written df or ν. For a single sample, df = n − 1. Small samples give small df and the heaviest tails.
- **Approaches the normal.** As n, and therefore df, grows, s becomes a dependable estimate of σ, the tails thin out, and the t curve approaches the standard normal curve. By about df = 30 the two are close, which is why z and t procedures merge for large samples.

- **Mean zero.** The distribution is centred at zero, the value the statistic takes when the sample mean exactly equals the hypothesised population mean.

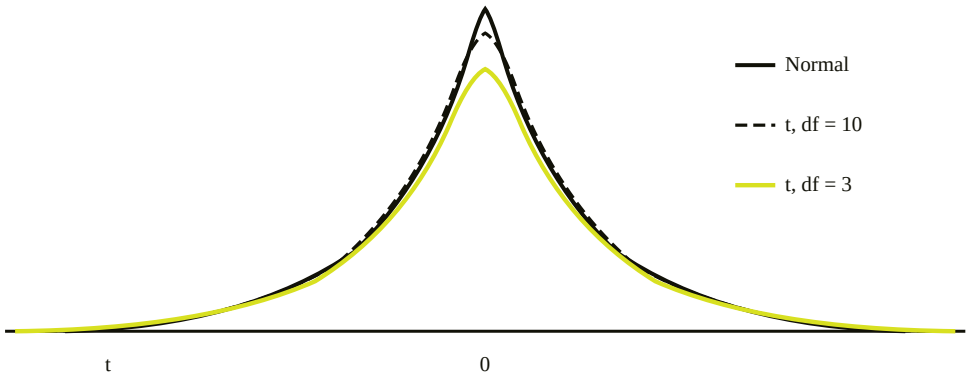


Figure 1 — t curves for df = 3 and df = 10 compared with the normal curve: heavier tails for small df, approaching the normal as df grows

Figure 1 is a picture of uncertainty: with few observations the price of estimating σ is a visibly flatter curve, and the test becomes more cautious automatically. As df rises the caution is withdrawn until, for large samples, the normal critical value 1.96 of Topic 9 is recovered.

3. Conditions and assumptions

The one-sample t-test is valid under the following conditions, which should be checked, or at least considered, before the arithmetic begins.

- The observations form a **random sample** of independent measurements from the population of interest.
- The **population is normal**, or approximately normal. This assumption matters most for very small samples; moderate departures from normality are tolerated reasonably well, especially for two-tailed tests, but strong skewness or extreme outliers can mislead the test.
- The sample is **small** in the usual working sense, and the population standard deviation σ is **unknown**, so it is estimated by the sample standard deviation s, computed with divisor n − 1.
- The variable is measured on a continuous scale, as with breaking load, length, mass or temperature in textile testing.

Where the population is clearly far from normal and the sample is tiny, the honest course is to consider a transformation or a distribution-free method, rather than report a t value whose table probabilities do not apply.

4. The one-sample t-test

To test H_0: $\mu = \mu_0$ against a one-tailed or two-tailed alternative, compute

$$t = (\bar{x} - \mu_0) / (s / \sqrt{n}), \quad \text{with } df = n - 1$$

where $\bar{x}$ is the sample mean, s is the sample standard deviation with divisor $n - 1$, and n is the sample size. The denominator $s / \sqrt{n}$ is the estimated standard error of the mean. The calculated $|t|$ is compared with the table value of t at the chosen level for $df = n - 1$, exactly as z was compared with 1.96 in Topic 9. If $|t|$ exceeds the table value, reject H_0; otherwise do not reject it.

The structure, (observed value minus hypothesised value) divided by standard error, is the same as for z. What changes is the yardstick: the table value depends on the degrees of freedom and, for small samples, is noticeably larger than 1.96, reflecting the heavier tails of Figure 1.

5. Fully worked example: a claim about mean breaking load

A supplier claims that the mean breaking load of a yarn is $\mu_0 = 64$ units. A random sample of $n = 10$ specimens gives the following breaking loads. Test the claim at the 5% level.

SPECIMEN	LOAD X	DEVIATION $(X - \bar{X})$	$(X - \bar{X})^2$
1	60	−6	36
2	64	−2	4
3	66	0	0
4	68	2	4
5	70	4	16
6	62	−4	16
7	65	−1	1
8	67	1	1
9	69	3	9
10	69	3	9
Total	**$\Sigma x = 660$**	**0**	**$\Sigma(x - \bar{x})^2 = 96$**

WORKED EXAMPLE

Hypotheses. H_0: $\mu = 64$; H_1: $\mu \neq 64$ (two-tailed; the claim fails if the true mean is either lower or higher). Level $\alpha = 0.05$.

Sample mean. $\bar{x} = \Sigma x / n = 660 / 10 = 66$.

Sample variance and standard deviation. $s^2 = \Sigma(x - \bar{x})^2 / (n - 1) = 96 / 9 = 10.67$, so $s = \sqrt{10.67} = 3.27$.

Test statistic. $t = (\bar{x} - \mu_0) / (s / \sqrt{n}) = (66 - 64) / (3.27 / \sqrt{10}) = 2 / 1.033 = 1.94$, with $df = n - 1 = 9$.

Critical value. From the t table, $t(0.05, 9\ df, \text{two-tailed}) = 2.262$.

Decision. $1.94 < 2.262$, so the calculated value does not reach the critical region. Accept H_0 at the 5% level: although the sample mean, 66, sits above the claimed 64, a sample of only 10 specimens varies enough that this excess is not significant. The sample does not disprove the supplier's claim.

Note the comparison with z: a standardised difference of 1.94 comes close to 1.96, yet the correct t critical value for 9 df is 2.262, comfortably larger. Using z here would have exaggerated the evidence; the t-test builds the smallness of the sample into the decision.

6. The paired t-test

Many comparisons are naturally **paired**: the same machines, looms or fabric pieces are measured before and after a change. Pairing removes the large variation between units from the comparison; what matters is the difference within each pair.

For each pair form the difference d (for example, after minus before). Then test whether the mean difference is zero using the ordinary one-sample formula applied to the differences:

$$t = \bar{d} / (s_d / \sqrt{n}), \quad \text{with } df = n - 1$$

where $\bar{d}$ is the mean of the differences, s_d is their standard deviation with divisor $n - 1$, and n is the number of pairs. H_0 states that the true mean difference is zero, that is, that the change has had no effect.

WORKED EXAMPLE

Five machines are measured before and after an adjustment. The gains in output (after minus before) are d = 2, 3, 1, 4, 2. Test at the 5% level whether the adjustment has improved output.

Mean difference. $\bar{d} = (2 + 3 + 1 + 4 + 2) / 5 = 12 / 5 = 2.4$.

Deviations from $\bar{d}$. $-0.4, 0.6, -1.4, 1.6, -0.4$; their squares are 0.16, 0.36, 1.96, 2.56, 0.16, giving $\Sigma(d - \bar{d})^2 = 5.20$.

Standard deviation of differences. $s_d^2 = 5.20 / 4 = 1.30$, so $s_d = 1.14$.

Test statistic. $t = \bar{d} / (s_d / \sqrt{n}) = 2.4 / (1.14 / \sqrt{5}) = 2.4 / 0.510 = 4.71$, with $df = 5 - 1 = 4$.

Critical value and decision. The table value $t(0.05, 4\ df) = 2.776$. Since $4.71 > 2.776$, reject H_0 at the 5% level: the improvement in output after the adjustment is statistically significant, and is larger than ordinary variation between repeated measurements would explain.

Treating paired data as two independent samples wastes the design: machine-to-machine differences would inflate the standard error and could hide a real, consistent gain. Whenever data arrive as matched pairs, analyse the differences.

7. Two independent samples: the pooled t-test

When two independent samples, of sizes n_1 and n_2, come from normal populations with a common (but unknown) variance, the difference of the means is tested with the **pooled** estimate of variance:

$$s^2 = [\ (n_1 - 1)s_1^2 + (n_2 - 1)s_2^2\ ] / (n_1 + n_2 - 2)$$

$$t = (\bar{x}_1 - \bar{x}_2) / [\ s\ \sqrt{(1/n_1 + 1/n_2)}\ ], \quad \text{with } df = n_1 + n_2 - 2$$

Pooling weights each sample variance by its degrees of freedom, so the larger sample contributes more. H_0 usually states that the two population means are equal. If the two populations clearly have very different spreads, the pooled form is not the right tool and an unpooled variant is preferred. The computed $|t|$ is then compared with the table value at $df = n_1 + n_2 - 2$.

8. Uses and limitations

Uses. The t-test is the everyday instrument for small-sample questions about a mean: checking a supplier’s claim from a handful of specimens, judging whether a process change has shifted a mean, comparing two small groups, and analysing before-and-after trials through pairing. In textile testing, where specimens are costly and batches are small, it is arguably the most frequently used test of all.

Limitations. The normality assumption matters more as the sample shrinks; with n as small as 5, one outlier can dominate both $\bar{x}$ and s. The test concerns the mean only: two populations can share a mean while differing in spread or shape, undetected. It reports statistical significance, not practical importance — with a large sample a trivial difference in breaking load can become “significant” yet remain of no consequence for the fabric. Finally, it requires genuinely independent observations (or genuine pairs); repeated readings on one specimen are not independent specimens, and treating them as such invents degrees of freedom that do not exist.

Glossary

TERM	MEANING
Student’s t distribution	The sampling distribution of $(\bar{x} - \mu) / (s / \sqrt{n})$ for a small sample from a normal population with σ unknown.
Degrees of freedom (df)	The index of a t curve; for one sample df = n − 1, the number of deviations free to vary once the mean is fixed.
Heavier tails	The greater probability of extreme values under t than under the normal curve, reflecting uncertainty in s.
Sample standard deviation (s)	The estimate of σ computed with divisor n − 1: $s^2 = \Sigma(x - \bar{x})^2 / (n - 1)$.
One-sample t-test	The test of H_0: $\mu = \mu_0$ using $t = (\bar{x} - \mu_0) / (s / \sqrt{n})$ with df = n − 1.
Paired t-test	A one-sample t-test applied to the differences within matched pairs, testing whether the mean difference is zero.
Pooled variance	The combined estimate of a common variance from two samples, weighted by their degrees of freedom.
Critical value of t	The table value, for given df and level, that the computed |t| must exceed for H_0 to be rejected.

Key points & summary

- Use t, not z, when the sample is small, σ is unknown and must be estimated by s, and the population is approximately normal.
- The t curve is symmetric with heavier tails than the normal curve; as df grows it approaches the normal curve.
- One-sample test: $t = (\bar{x} - \mu_0) / (s / \sqrt{n})$, df = n − 1; compare |t| with the table value for that df.
- In the breaking-load example, $\bar{x} = 66$, $s = 3.27$ and $t = 1.94 < 2.262$ (9 df, 5%), so the claim $\mu = 64$ is not disproved.
- For paired data, analyse the differences: $t = \bar{d} / (s_d / \sqrt{n})$, df = number of pairs minus 1; in the machine example $t = 4.71 > 2.776$, so the improvement is significant.
- For two independent samples with a common variance, pool the variances and use $df = n_1 + n_2 - 2$.
- Statistical significance is not practical importance, and the t-test judges the mean only, not spread or shape.

Practice

Answer in your own words; brief model answers follow.

1. Give two reasons why the z-test is unsuitable for testing a mean breaking load from a sample of 8 specimens when σ is unknown.
2. Sketch, in words, how the t curve for df = 3 differs from the normal curve, and what happens to the curve as df increases to 30.
3. A sample of n = 16 has $\bar{x} = 48$ and s = 6. Compute t for testing H_0: $\mu = 50$, state the degrees of freedom, and say what further information you need to reach a decision at the 5% level.
4. In the breaking-load worked example, explain why the table value 2.262 rather than 1.96 is the correct standard of comparison, and what would have been wrong with using 1.96.
5. A paired experiment gives differences d = 1, 2, 2, 3, 2 for 5 pairs. Compute $\bar{d}$ and, given $s_d = 0.71$, compute t and outline the decision procedure at the 5% level (table value 2.776).
6. State the assumptions behind the pooled two-sample t-test, and one situation in textile testing where pairing is preferable to two independent samples.

Answer key

1. The sample is too small for the central limit theorem to guarantee a normal sampling distribution of the mean, and σ must be estimated by s from the same small sample, adding uncertainty that z ignores. The t distribution allows for both.
2. The df = 3 curve has the same centre and symmetry but a lower peak and much heavier tails, so extreme values are more probable. As df increases the peak rises, the tails thin, and by df = 30 the curve is close to the standard normal curve.
3. $t = (48 - 50) / (6 / \sqrt{16}) = -2 / 1.5 = -1.33$, so $|t| = 1.33$, with df = 15. To decide, the table value of t at the 5% level for 15 df is needed; since t critical values for moderate df lie near 2.1, $|t| = 1.33$ would not lead to rejection of H_0.
4. Because s is estimated from only 10 observations, the statistic follows t with 9 df, whose 5% two-tailed critical value is 2.262. Using 1.96 would apply the normal standard, understate the uncertainty in s, and risk rejecting H_0 on evidence that is not strong enough — a raised Type I error rate.
5. $\bar{d} = 10 / 5 = 2.0$. $t = 2.0 / (0.71 / \sqrt{5}) = 2.0 / 0.318 = 6.30$. Since $6.30 > 2.776$, reject H_0 at the 5% level: the mean difference is significantly different from zero.
6. Both populations normal, a common population variance, and independent random samples. Pairing is preferable when the same looms or fabric pieces can be measured before and after a treatment, because it removes loom-to-loom or piece-to-piece variation from the comparison.